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Truthful reverse auction-based incentive mechanisms for task offloading in mobile edge computing

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ABSTRACT

Offloading computation-intensive tasks of mobile devices (MDs) to the nearby base stations (BSs) equipped with edge servers has been a promising approach to effectively alleviate the problem of insufficient computation capacity of MDs. Nevertheless, task offloading needs to consume numerous resources (e.g., computation and communication resources, etc.). Assumed that the owners of BSs are selfish and rational, they will be reluctant to take part in the task offloading without acquiring suitable economic reimbursement. Therefore, it is necessary to design an effective incentive mechanism to motivate BSs to engage in the process of task offloading. In this paper, we present a truthful reverse auction-based incentive mechanism (TRAIM) with explicit consideration of the locality constraint, overlapped coverage and capacity constraint to offload tasks remotely. To be specific, we first devise the *MD assignment* based on the dynamic programming approach to acquire the set of optimal BS-MD associations. Next, we devise the *winning BS selection* using the greedy method to decide the set of winning BSs. Then, we compute the charge of each winning BS according to its critical value, aiming to prevent the strategic behaviors. Strict theoretical analysis demonstrates our proposed TRAIM is truthful, individually rational and computationally efficient. Extensive simulations also demonstrate the effectiveness of our proposed TRAIM.

1. Introduction

With the exponential increase of mobile devices (MDs) and the rapid development of wireless communications, an increasing number of mobile applications with advanced features, such as face recognition [1], real-time online gaming [2] and virtual/augmented reality [3], are emerging to facilitate user experiences. Nevertheless, such complicated applications are often computation-intensive, which bring significant challenges to MDs with finite computation resources due to their constrained physical size. In fact, offloading these complicated applications to the centralized cloud for remote execution can effectively alleviate the problem of insufficient computation capacity of MDs. Nevertheless, this way may lead to inevitably long response delays, since clouds are often located far from MDs, which reduces the service quality of these mobile applications with strict timeliness constraints.

To solve the aforementioned challenge, mobile edge computing (MEC) is regarded as a new promising computing paradigm for enabling computation-intensive mobile applications [4]–[7]. In MEC, many base stations (BSs) equipped with edge servers are deployed in the vicinity of MDs, and each MD can offload the computation-intensive task to the nearby BSs for remote execution. As such, the service quality of MDs will be significantly improved. However, the process of task offloading inevitably incurs the consumptions of computation and subchannel resources from BSs. As a result, they may be unwilling to take part in the process of task offloading without acquiring the suitable economic reimbursement. Consequently, it is urgent to design effective incentive mechanisms to motivate BSs to engage in the task offloading process. Since auction has been widely applied in incentive mechanism design in different fields, e.g., mobile crowdsensing [8], smart grid [9] and network function virtualization [10], we study the incentive mechanism for task offloading based on the reverse auction, where each bidder (as a seller) offers items to the user (as a buyer).

Nevertheless, designing an effective reverse auction-based incentive mechanism for task offloading is extremely complicated because of

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BSs' strategic behavior issues [11]–[14]. Strategic BSs are usually self-interested and selfish. Thus, BSs may tell the untruthful costs in order to improve the utilities regardless of other BSs' utilities. Therefore, in the long term, the MEC system may suffer from significant economic losses. What is more, the cost information is private and not released to MDs, who cannot access the real costs. As a result, in an MEC incentive mechanism, it is difficult for MDs to trust the cost information supplied by BSs and MDs need to act cautiously due to the possibility of information falsification by BSs. In this context, a promising research direction is mechanism design [15]–[17], which explores how the truthful and private information such as the real cost can be revealed. Consequently, designing a truthful reverse auction-based incentive mechanism is crucial to encourage BSs to report the costs honestly.

In reality, when striving to ensure truthfulness of BSs, the characteristics of task offloading in an MEC system also need to be considered. Typically, BSs in MEC systems are geo-distributed, each of which only covers a specific geographical area wirelessly because of the finite transmission power. As a result, an MD can only offload the task to the BSs which cover it and a BS can only serve MDs which are in its coverage, known as locality constraint. Moreover, we also need to consider the overlapped coverage, that is, the same MD may appear in the overlapping region of BSs. In addition, a BS may cover a large number of MDs simultaneously, and the computation and subchannel resources of BSs are finite, known as capacity constraint.

However, the above characteristics of task offloading increases the difficulty of achieving truthfulness of BSs. In fact, a reverse auction-based incentive mechanism for task offloading mainly involves the following tasks: *1) How to design the suitable scheme for task offloading and resource allocation to acquire low-complexity solutions? 2) How to encourage BSs to take part in offloading tasks and how to calculate the charges of the winning BSs? 3) How to minimize the social cost while meeting the specific constraints?* Our goal is to minimize social cost and achieve truthfulness in the above tasks, while explicitly considering the characteristics of task offloading.

Consequently, we present a truthful reverse auction-based incentive mechanism (TRAIM) for task offloading in an MEC system. Differing from existing researches, TRAIM is applicable to the reverse auction scenario, where each bidder (as a seller) offers items to the user (as a buyer). Furthermore, we explicitly take into account the characteristics of task offloading, such as locality constraint, overlapped coverage and capacity constraint, etc., which significantly complicates the design of TRAIM. To be specific, we first devise the *MD assignment* based on the dynamic programming approach to obtain the set of optimal BS-MD associations. Next, we devise the *winning BS selection* with the greedy method to decide the set of winning BSs in polynomial time. Then, we compute the charges of each winning BS according to its critical value, aiming to prevent the strategic behaviors.

The main contributions of this paper are as follows:

- To the best of our knowledge, our proposed TRAIM is the first truthful reverse auction-based incentive mechanism with explicit consideration of the locality constraint, overlapped coverage and capacity constraint in the process of MEC task offloading.
- We propose TRAIM which is primarily composed of the *MD assignment*, the *winning BS selection* and the *critical-value-based charging*. In the *MD assignment*, we design a dynamic programming based algorithm to obtain the set of optimal BS-MD associations. In the *winning BS selection*, we devise a greedy algorithm to decide the collection of winning BSs. In the *critical-value-based charging*, we compute the charge of each winning BS based on its critical value.
- Strict theoretical analysis demonstrates our proposed TRAIM is truthful, individually rational and computationally efficient. Extensive simulations also demonstrate the effectiveness of our proposed TRAIM.

In the rest of this paper, we describe the system model and formally introduce the problem in Section 2. Section 3 presents our proposed TRAIM. The theoretical analysis is conducted in Section 4. Section 5 evaluates the performance of our proposed TRAIM. The related works are reviewed in Section 6. Section 7 concludes this paper.

2. Preliminary and problem formulation

We first present the system model that depicts different characteristics of task offloading process between MDs and BSs. Then, we define the task offloading as mechanism design, targeting to guarantee truthfulness, individual rationality of BSs, and obtain an approximately optimal solution with explicit consideration of the locality constraint, overlapped coverage and capacity constraint. The key notations are summarized in Table 1.

2.1. System model

We consider an MEC system involving a platform, multiple MDs and multiple BSs. The platform receives task offloading requests from MDs and periodically publicizes these task offloading requests to be executed by BSs. Let $I = \{1, 2, \ldots, |I|\}$ be the set of MDs. The request of each MD $i \in I$ is represented as $R_i = \{(\chi_i, \gamma_i), h_i, d_i\}$, where (χ_i, γ_i) denotes the longitude and latitude of MD i, h_i denotes the required data transmission rate of MD i, d_i denotes the required number of CPU cores of MD i. Note that the required data transmission rate and the required number of CPU cores can be computed by an MD based on its deadline, which has been investigated in [18],[19] and is orthogonal to the focus of our work.

Let $J = \{1, 2, \ldots, |J|\}$ denote the collection of BSs. The bid of each BS $j \in J$ is represented as $B_j = \{(\alpha_j, \beta_j), F_j, \theta_j, \omega_j, \delta_j, b_j\}$, where (α_j, β_j) denotes the longitude and latitude of BS j, F_j represents the available number of CPU cores of BS j, θ_j denotes the available number of orthogonal wireless subchannels of BS j, ω_j denotes the bandwidth of one subchannel, δ_j is the maximum coverage radius of BS j, and b_j is the claimed cost of these computation and subchannel resource combinations. Noting that, following the locality constraint, BS j can only server the MDs with a circular region centered at (α_j, β_j) with radius δ_j.

Table 1
Key notations.

Notation	Description
I	set of MDs
R_i	request of each MD i
d_i	required number of CPU cores of MD i
h_i	required data transmission rate of MD i
(χ_i, γ_i)	longitude and latitude of MD i
J	set of BSs
B_j	bid of BS j
(α_j, β_j)	longitude and latitude of BS j
F_j	available number of CPU cores of BS j
θ_j	available number of orthogonal wireless subchannels of BS j
ω_j	bandwidth of one subchannel
δ_j	maximum coverage radius of BS j
b_j	claimed cost
x_j	whether or not accept bid B_j
y_{ij}	whether or not choose BS j to execute MD i's task
p_j	charge of BS j
u_j	utility of BS j
$V(J)$	social cost
I_j	set of MDs in BS j's coverage region
J_i	set of BSs which cover MD i
$r(B_j)$	CRP of bid B_j
$C(B_j)$	critical bid of bid B_j
W	set of winning BSs
L_W	set of served tasks
T_j	set of optimal BS-MD associations of BS j
ϕ_{ij}	number of subchannels of MD i leasing from BS j
$\varphi(z)$	index of MD in the zth position of the sort
Q_j	number of MDs in I_j
p_j^{CV}	critical value of BS j

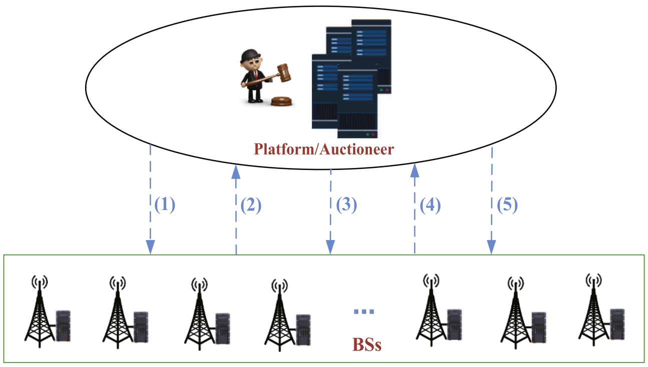

Fig. 1. The interactions between a platform and BSs in the reverse auction framework.

2.2. Reverse auction framework

In an MEC system, BSs compete to get the chances to offer resources to offload MDs' tasks. The interactions between a platform and BSs are modeled as a reverse auction framework, where the platform acts as the auctioneer and the buyer (acquiring resources for task offloading) and BSs act as both sellers and bidders (offering resources). The reverse auction framework mainly involved five steps, as shown in Fig. 1.

1) The platform publicizes MDs' requests to all the BSs in the MEC system.

2) Each BS j submits bid B_j to the platform. Let x_j denote a binary variable, which indicates whether bid B_j is accepted ($x_j = 1$) or not ($x_j = 0$). Let y_{ij} be a binary variable, which indicates whether BS j is chosen to execute MD i's task ($y_{ij} = 1$) or not ($y_{ij} = 0$).

3) The platform calculates the allocation profiles, i.e., $\{x_j\}_{j\in J}$ and $\{y_{ij}\}_{i\in I, j\in J}$, decides the charge p_j of each BS, and announces the results to BSs.

4) Each winning BS j executes its corresponding tasks and sends the results to the platform.

5) Each winning BS j receives an amount of money p_j from the platform for supplying resources to execute task offloading.

In addition, since each BS j is selfish and rational, its claimed cost b_j may be more or less than its real cost c_j, in order to improve its utility.

2.3. Problem formulation

We first introduce a number of necessary definitions, and define the resource allocation problem for task offloading (RAPTO).

We denote by $B = (B_1, \dots, B_{|J|})$ the vector of bids of all BSs. Besides, B_{-j} is the vector of all bids except BS j's bid (i.e., $B_{-j} = (B_1, \dots, B_{j-1}, B_{j+1}, \dots, B_{|J|})$).

Definition 1. (BS Utility). BS j's utility equals its charge minus its real cost, i.e.,

$$u(B_j, B_{-j}) = p_j - c_j \tag{1}$$

Definition 2. (Social Cost). Social cost equals the summation of the true costs of the winning bids, i.e.,

$$V(J) = \sum_{j\in J} x_j c_j \tag{2}$$

Definition 3. (Truthfulness). A mechanism achieves truthfulness if for each BS j, for each submitted bid of the other BSs B_{-j}, a true bid $\tilde{B}_j = \{(\alpha_j, \beta_j), F_j, \theta_j, \omega_j, \delta_j, c_j\}$ and any other submitted bid $B_j = \{(\alpha_j, \beta_j), F_j, \theta_j, \omega_j, \delta_j, b_j\}$ of BS j, we have $u(\tilde{B}_j, B_{-j}) \ge u(B_j, B_{-j})$.

Definition 4. (Individual Rationality). A mechanism achieves individual rationality if for each BS j claiming its true bid $\tilde{B}_j = \{(\alpha_j, \beta_j), F_j, \theta_j, \omega_j, \delta_j, c_j\}$, we have $u(\tilde{B}_j, B_{-j}) \ge 0$, for all other BSs' bids B_{-j}.

Definition 5. (Computational Efficiency). A mechanism achieves computational efficiency if it finishes in polynomial time.

Under bidding truthfully, the RAPTO for social cost minimization is defined as an integer linear program as follows:

$$\min V(J) = \sum_{j\in J} x_j c_j \tag{3}$$

$$\sum_{i\in I_j} y_{ij} d_i \le F_j, \quad \forall j \in J \tag{3.1}$$

$$\sum_{i\in I_j} y_{ij} \phi_{ij} \le \theta_j, \quad \forall j \in J \tag{3.2}$$

$$\sum_{j\in J_i} y_{ij} \le 1, \quad \forall i \in I \tag{3.3}$$

$$y_{ij} \le x_j, \forall i \in I, j \in J_i \tag{3.4}$$

$$x_j \in \{0, 1\}, \forall j \in J \tag{3.5}$$

$$y_{ij} \in \{0, 1\}, \forall i \in I, j \in J_i \tag{3.6}$$

$$I_j = \{i | i \in I, \sqrt{(\chi_i - \alpha_j)^2 + (\gamma_i - \beta_j)^2} \le \delta_j\}, \forall j \in J \tag{3.7}$$

$$J_i = \{j | j \in J, \sqrt{(\chi_i - \alpha_j)^2 + (\gamma_i - \beta_j)^2} \le \delta_j\}, \forall i \in I \tag{3.8}$$

In a truthful auction, $b_j = c_j$. Constraint (3.1) guarantees that for each BS, the consumption of CPU cores should not exceed its total CPU cores. Constraint (3.2) ensures that for each BS, the consumption of subchannels should not exceed its total number of subchannels, where ϕ_{ij} is the number of subchannels of MD i leasing from BS j, which can satisfy the requirement of data transmission rate h_i. Constraint (3.3) guarantees that each MD's task can be offloaded by one BS only in order to avoid overlapping. Constraint (3.4) ensures that only the BS who wins the auction can execute task offloading. Constraints (3.5) and (3.6) ensure the integer nature of binary decision variables. Constraints (3.7) and (3.8) guarantee the locality requirement, I_j denotes the set of MDs in BS j's coverage region and J_i denotes the set of BSs which cover MD i.

It is rather difficult to resolve the RAPTO which involves integer constraints $x_j \in \{0, 1\}$, $y_{ij} \in \{0, 1\}$ and division mathematical operation of these variables. Compared with the 0–1 knapsack problem, the RAPTO is more complex and belongs to an NP-hard problem. Because no algorithm can resolve the RAPTO in polynomial time, we design an heuristic algorithm to calculate the approximate optimal solution, while achieving the important properties.

3. Design of TRAIM

In this section, we present the detials of TRAIM, which mainly includes the *MD assignment*, the *winning BS selection* and the *critical-value-based charging*. In the *MD assignment*, we design a dynamic programming based algorithm to calculate the set of optimal BS-MD associations. In the *winning BS selection*, we devise a greedy algorithm to decide the winning BS set. In the *critical-value-based charging*, we compute the charges of winning BSs using their corresponding critical values.

3.1. MD assignment

For each BS j, we design a dynamic programming based algorithm to find the set of optimal BS-MD associations T_j in its coverage region, aiming to maximize its consumed normalized resources.

First, for each BS $j \in J$, we sort the set of MDs in its coverage region, i.e., I_j, in an ascending order of i. Then, we have MD $\varphi(z) \in I_j$ as follows:

$$\varphi(1) < \varphi(2) < \dots < \varphi(Q_j) \tag{4}$$

Here, $\varphi(z)$ denotes the index of MD in the zth position of the sort, and $Q_j = |I_j|$ denotes the number of MDs in I_j.

Next, given that the available CPU cores and subchannels of BS j are respectively F_j and θ_j, we determine which MDs in I_j can maximize the consumed normalized resources of BS j.

Let $N_j[z,\mu,\lambda]$ denote the maximum consumed normalized resources of BS j if MD set $\{MD\ \varphi(1),\dots,MD\ \varphi(z)\}$ are in its coverage region and its available CPU cores are $\mu\in[0,F_j]$ and its available subchannels are $\lambda\in[0,\theta_j]$. Thus, the state transition equation can be defined as:

$$N_j[z,\mu,\lambda]=\max\left\{N_j[z-1,\mu,\lambda],\right.$$
$$\left.N_j[z-1,\mu-d_{\varphi(z)},\lambda-\phi_{\varphi(z)j}]+\frac{d_{\varphi(z)}}{F_{\max}}+\frac{\phi_{\varphi(z)j}}{\theta_{\max}}\right\} \tag{5}$$

where $F_{\max}=\max_{j\in J}\{F_j\}$, $\theta_{\max}=\max_{j\in J}\{\theta_j\}$, and $\phi_{\varphi(z)j}$ is the number of subchannels of MD $\varphi(z)$ obtaining from BS j such that MD $\varphi(z)$ can satisfy the requirement of data transmission rate $h_{\varphi(z)}$. Then, $\phi_{\varphi(z)j}$ can be computed by:

$$\phi_{\varphi(z)j}=\left\lceil\frac{h_{\varphi(z)}}{\omega_j\log_2\left(1+\eta_{\varphi(z)}\sigma_{\varphi(z)j}/\varpi_{\varphi(z)}\right)}\right\rceil \tag{6}$$

where $\eta_{\varphi(z)}$ denotes the transmission power of MD $\varphi(z)$, $\sigma_{\varphi(z)j}$ denotes the channel gain between BS j and MD $\varphi(z)$, and $\varpi_{\varphi(z)}$ represents the background noise.

By (5), if MD $\varphi(z)$ is not accepted by BS j, then $N_j[z,\mu,\lambda]=N_j[z-1,\mu,\lambda]$; if MD $\varphi(z)$ is accepted by BS j, then $N_j[z,\mu,\lambda]=N_j[z-1,\mu-d_{\varphi(z)},\lambda-\phi_{\varphi(z)j}]+d_{\varphi(z)}/F_{\max}+\phi_{\varphi(z)j}/\theta_{\max}$. As a result, the maximum consumed normalized resources of BS j is $N_j[Q_j,F_j,\theta_j]$, which can be acquired through traversing the state of each MD $\varphi(z)\in I_j$ based on the state transition equation. What is more, we can further acquire the set of optimal BS-MD associations T_i according to the state transition of the final state $N_j[Q_j,F_j,\theta_j]$. More specifically, in Algorithm 1, we first calculate the values of $N_j[1,\mu,\lambda]$, for $\mu\in\{1,2,\dots,F_j\}$ and $\lambda\in\{1,2,\dots,\theta_j\}$ (Lines 4–12). Next, according to the state transition Eq. (5), we calculate the values of $N_j[z,\mu,\lambda]$, for $z\in\{2,3,\dots,Q_j\}$, $\mu\in\{1,2,\dots,F_j\}$ and $\lambda\in\{1,2,\dots,\theta_j\}$ (Lines 13–22). Thus, we can obtain the value of $N_j[Q_j,F_j,\theta_j]$, i.e., the maximum consumed normalized resources of BS j if MD set $\left\{MD\ \varphi(1),\dots,MD\ \varphi(Q_j)\right\}$ are in its coverage region and its available CPU cores are F_j and its available subchannels are θ_j. After obtaining the value of $N_j[Q_j,F_j,\theta_j]$, we use the backtracking method to determine these accepted MDs, according to the condition $N_j[z,\mu,\lambda]=N_j[z-1,\mu-d_{\varphi(z)},\lambda-\phi_{\varphi(z)j}]+d_{\varphi(z)}/F_{\max}+\phi_{\varphi(z),j}/\theta_{\max}(Lines24-29)$.

Algorithm 1. MD assignment.

Input: I, J, $\{R_i\}$, B_j.
Output: T_j.
1: $T_j=\emptyset$;
2: $I_j=\{i\mid i\in I,\sqrt{(\chi_i-\alpha_j)^2+(\gamma_i-\beta_j)^2}\le\delta_j\}$;
3: After sorting MD $i\in I_j$ according to i in an ascending order, let $\varphi(z)$ denote the index of MD at the position z in the ordering and the last one in the ordering is Q_j;
4: **for** $\mu\leftarrow 1$ to F_j **do**
5: **for** $\lambda\leftarrow 1$ to θ_j **do**
6: **if** $\mu\ge d_{\varphi(1)}\wedge\lambda\ge\phi_{\varphi(1)j}$ **then**
7: $N_j[1,\mu,\lambda]=d_{\varphi(1)}/F_{\max}+\phi_{\varphi(1)j}/\theta_{\max}$;
8: **else**
9: $N_j[1,\mu,\lambda]=0$;
10: **end if**
11: **end for**
12: **end for**
13: **for** $z\leftarrow 2$ to Q_j **do**
14: **for** $\mu\leftarrow 1$ to F_j **do**
15: **for** $\lambda\leftarrow 1$ to θ_j **do**
16: $N_j[z,\mu,\lambda]=N_j[z-1,\mu,\lambda]$;
17: **if** $\mu\ge d_{\varphi(z)}\wedge\lambda\ge\phi_{\varphi(z)j}$ **then**
18: Calculate $N_j[z,\mu,\lambda]$ according to (5);
19: **end if**
20: **end for**
21: **end for**
22: **end for**
23: $z'=Q_j,\mu'=F_j,\lambda'=\theta_j$;
24: **while** $z'>0$ **and** $\mu'>0$ **and** $\lambda'>0$ **do**
25: **if** $N_j[z',\mu',\lambda']=N_j[z'-1,\mu'-d_{\varphi(z')},\lambda'-\phi_{\varphi(z')j}]+d_{\varphi(z')}/F_{\max}+\phi_{\varphi(z')j}/\theta_{\max}$ **then**
26: $T_j\leftarrow T_j\cup\varphi(z')$;
27: **end if**
28: $z'=z'-1$, $\mu'=\mu'-d_{\varphi(z')}$, $\lambda'=\lambda'-\phi_{\varphi(z')j}$;
29: **end while**
30: **return** T_j;

3.2. Winning BS selection

Due to the NP-hardness of the RAPTO, it is impossible to find the optimal solution within polynomial time. A promising way to solve the RAPTO is to design a heuristic approximation approach. In general, an approximation approach may lack the necessary properties for ensuring truthfulness, and thereby, it requires to be deliberately designed to achieve truthfulness. Therefore, we design a truthful greedy approximation algorithm to solve the RAPTO, i.e., the *winning BS selection*. For ease of understanding, we first assume that BSs submit the truthful bids, and next demonstrate that BSs would be truthful in the following section.

The main idea of the *winning BS selection* is to repeatedly choose the next most cost-efficient bid that makes progress towards offloading all the tasks until all the tasks are offloading. That is, the bid which has the minimum cost per consumed normalized resource (CPR) will be accepted in each iteration. In specific, for each BS j, its related CPR, i.e., $r(B_j)$, can be computed as follows:

$$r(B_j)=\frac{b_j}{\sum_{i\in T_j}\left(d_i/F_{\max}+\phi_{ij}/\theta_{\max}\right)} \tag{7}$$

The *winning BS selection* is described as Algorithm 2. It has four input parameters: I, J, R_i, B_j, and one output parameter: W (see Table 1 for details). The *winning BS selection* in each iteration works as follows. First, for each BS j, if I_j equals $\emptyset$, it will be deleted from M (Lines 4–5); otherwise, T_j will be calculated using Algorithm 1, and the corresponding CPR will further be computed (Lines 7–8). Next, the bid who has the minimum CPR is picked as the winning bid (Line 11), and the corresponding variables, like M, W and L_W are updated (Lines 12–14). Finally, for each remainder BS j, the corresponding set I_j are also updated (Lines 15–17).

3.3. Critical-value-based charging

Once determining the winning BSs, the platform needs to calculate their charges. The charge of each winning BS requires to be decided in such a way that guarantees BSs tell their costs truthfully. Therefore, we employ the critical-value strategy to calculate the charge of each BS.

Each winning BS j would charge a certain amount of monetary compensation for the consumed resources. The amount is calculated based on its critical bid $C(B_j)$ that can be determined as follows: if bid B_j satisfies $r(B_j)\le r(C(B_j))$ it will be accepted, otherwise it will be rejected if $r(B_j)>r(C(B_j))$. The critical bid $C(B_j)$ is the first bid that renders bid B_j to be rejected. Bid B_j is rejected when it cannot offload any task. Because the charge of each winning bid B_j is related to the claimed cost of $C(B_j)$, the charge is called critical-value-based charge.

In specific, for each winning bid B_j, the method to determine the critical bid is described as Algorithm 3. It has six input parameters: I, J, R_i, B_j, W and T_j, and one output parameter: $\{p_j\}$ (see Table 1 for details). In Algorithm 3, we first set p_j and p_j^{CV} to 0 and 0 (Line 1), where p_j^{CV}

Algorithm 2. Winning BS selection.

Input: I, J, R_i, B_j.
Output: W.
$W \leftarrow \emptyset, L_W \leftarrow \emptyset, M = J$;
while $L_W \neq I$ **do**
  for each $j \in M$ **do**
    if $I_j = \emptyset$ **then**
      $M = M \backslash \{j\}$;
    else
      calculate T_j using Algorithm 1;
      calculate $r(B_j)$ according to (7);
    end if
  end for
  $j' \leftarrow argmin_{j \in M}\{r(B_j)\}$;
  $M = M \backslash \{j'\}$;
  $W = W \cup \{j'\}$;
  $L_W = L_W \cup T_j$;
  for each $j \in M$ **do**
    $I_j = I_j \backslash L_W$;
  end for
end while
return W;

Algorithm 3. Critical-value-based charging.

Input: I, J, R_i, B_j, W, T_j.
Output: $\{p_j\}$.
1: $\forall j \in W,\ p_j = 0,\ p_j^{CV} = 0$;
2: **for each** $j \in W$ **do**
3:   $W' = \emptyset, L_{W'} = \emptyset, M' = J \backslash \{j\}$;
4:   **while** $L_{W'} \neq I$ **do**
5:     **for each** $j' \in M'$ **do**
6:       **if** $I_{j'} = \emptyset$ **then**
7:         $M' = M' \backslash \{j'\}$;
8:       **else**
9:         calculate $T_{j'}$ according to Algorithm 1;
10:         calculate $r(B_{j'})$ according to (7);
11:       **end if**
12:     **end for**
13:     $k \leftarrow argmin_{j' \in M'}\{r(B_{j'})\}$;
14:     $M' = M' \backslash \{k\}, W' = W' \cup \{k\}, L_{W'} = L_{W'} \cup T_k$;
15:     **for each** $j' \in M'$ **do**
16:       $I_{j'} = I_{j'} \backslash L_{W'}$;
17:     **end for**
18:     **if** $I_j \subseteq L_{W'}$ **then**
19:       $p_j^{CV} = r(B_k) \cdot \sum_{i \in T_j} \left(d_i/F_{\max} + \phi_{ij}/\theta_{\max}\right)$;
20:       $p_j = p_j^{CV}$;
21:       **break**
22:     **end if**
23:   **end while**
24: **end for**
25: **return** $\{p_j\}$;

denotes the critical value of BS j. Next, for each BS j, we delete j from J to construct M' (Line 3), and then repeatedly select other BSs until BS j is unnecessary, i.e., $I_j \subseteq L_{W'}$ (Lines 4–23). After determining its critical bid, i.e., B_k, the winning bid B_j charges an amount of money, which is proportional to the CPR of B_k, and the critical-value-based charge is computed according to $p_j^{CV} = r(B_k) \cdot \sum_{i \in T_j} \left(d_i/F_{\max} + \phi_{ij}/\theta_{\max}\right)$ (Line 19).

4. Theoretical analysis

In the section, we will prove that our proposed TRAIM is truthful, individually rational and computationally efficient through strict theoretical proofs.

4.1. Truthfulness

To demonstrate that TRAIM is truthful, we should guarantee that BSs should honestly report their true costs when other BSs' strategies hold unchanged. According to [20],[21], TRAIM achieves truthfulness if and only if it meets the conditions below. 1) The *winning BS selection* in TRAIM is monotonic; 2) The *critical-value-based charging* uses the critical value as the charge of each winning bid.

Before proving TRAIM satisfies the above two conditions, we first introduce several related definitions.

Definition 6. (Monotonicity). For any BS j, if bid $B_j = \{(\alpha_j, \beta_j), F_j, \theta_j, \omega_j, \delta_j, c_j\}$ is accepted, then bid $\tilde{B}_j = \{(\alpha_j, \beta_j), F_j, \theta_j, \omega_j, \delta_j, c_j - \varepsilon\}$ will also be accepted, where $\varepsilon > 0$.

Definition 7. (Critical Value). For each bid B_j, there exists a critical value η_j. When bid B_j tells a cost that is no more than η_j, it will be accepted; otherwise, it will be rejected.

Next, we will prove TRAIM achieves truthfulness through demonstrating that it satisfies the aforementioned conditions.

Lemma 1. *The winning BS selection in TRAIM is monotonic.*

Proof. A queue $Q = (\kappa^1, \kappa^2, \cdots)$ is created to record the accepted bids in the order they have been accepted. Suppose that bid B_j is accepted at the lth iteration and is substituted by bid $\tilde{B}_j = \{(\alpha_j, \beta_j), F_j, \theta_j, \omega_j, \delta_j, c_j - \varepsilon\}$, where $\varepsilon > 0$. By the definition of CPR, we get

$$r\left(\tilde{B}_j\right) = \frac{c_j - \varepsilon}{\sum_{i \in T_j} \left(d_i/F_{\max} + \phi_{ij}/\theta_{\max}\right)} \leq r\left(B_j\right) \tag{8}$$

where both $r\left(\tilde{B}_j\right)$ and $r(B_j)$ own the same denominator, i.e., $\sum_{i \in T_j} \left(d_i/F_{\max} + \phi_{ij}/\theta_{\max}\right)$. Thus, the *winning BS selection* should have accepted bid $\tilde{B}_j$ before or at the lth iteration. Therefore, the *winning BS selection* satisfies the condition of monotonicity. □

Lemma 2. *The critical-value-based charging in TRAIM charges winning bids according to their critical values.*

Proof. Assume that bid $B_j = \{(\alpha_j, \beta_j), F_j, \theta_j, \omega_j, \delta_j, p_j^{cv}\}$ is accepted with the resources $\sum_{i \in T_j} \left(d_i/F_{\max} + \phi_{ij}/\theta_{\max}\right)$, and the corresponding critical bid B_k is accepted at the lth iteration. Following the critical value charging scheme, we have $p_j^{cv} = \sum_{i \in T_j} \left(d_i/F_{\max} + \phi_{ij}/\theta_{\max}\right) \cdot r^l\left(B_k\right)$. It is obvious that bid $\tilde{B}_j = \{(\alpha_j, \beta_j), F_j, \theta_j, \omega_j, \delta_j, p_j^{cv} + \varepsilon\}$, $\varepsilon > 0$, whose cost is more than that of B_j will not be accepted, because

$$r^l\left(\tilde{B}_j\right) = \frac{p_j^{cv} + \varepsilon}{\sum_{i \in T_j} \left(d_i/F_{\max} + \phi_{ij}/\theta_{\max}\right)} > r^l\left(B_k\right) \tag{9}$$

which shows that the *winning BS selection* would accept the corresponding critical bid B_k. Furthermore, bid $\tilde{B}_j$ would not be accepted at the following iterations due to $I_j \subseteq L_W^l$. By contrast, bid $\tilde{B}_j = \{(\alpha_j, \beta_j), F_j, \theta_j, \omega_j, \delta_j, p_j^{cv} - \varepsilon\}$, $\varepsilon > 0$, whose claimed cost is less than that of B_j should be accepted, since at the lth iteration $r^l\left(\tilde{B}_j\right) < r^l\left(B_k\right)$. Consequently, the *winning BS selection* accepts bid $\tilde{B}_j$ with the less CPR and rejects the corresponding critical bid B_k. This demonstrates that p_j^{cv} is the critical value. □

Theorem 1. *TRAIM achieves truthfulness.*

Proof. According to Lemmas 1 and 2, we have the *winning BS selection* in TRAIM is monotone and the *critical-value-based charging* in TRAIM is the critical value charge. Consequently, by [20],[21], TRAIM achieves truthfulness. □

4.2. Individual rationality

Theorem 2. *TRAIM achieves individual rationality.*

Proof. For BS j, if its bid is not accepted, the utility is 0, i.e., $u_j = 0$. Conversely, for BS j, if its bid is accepted, we will demonstrate that the utility is nonnegative, e.g., $u_j \geq 0$. According to the charging strategy and the winning selection rule, we get $p_j^{cv} = \sum_{i \in T_j} \left(d_i/F_{\max} + \phi_{ij}/\theta_{\max}\right) \cdot r^l\left(B_k\right)$ and $b_j = \sum_{i \in T_j} \left(d_i/F_{\max} + \phi_{ij}/\theta_{\max}\right) \cdot r^l\left(B_j\right)$. Because TRAIM achieves truthfulness, we have $c_j = b_j$. Furthermore, $r^l\left(B_j\right) \leq r^l\left(B_k\right)$. Consequently, we get $p_j^{cv} \geq c_j$. As a result, TRAIM achieves individual rationality. □

4.3. Computational efficiency

Next, we will analyze the computational complexity of the *MD assignment*, the *winning BS selection* and the *critical-value-based charging* in TRAIM.

Lemma 3. *The MD assignment is computationally efficient.*

Proof. In each iteration, the outer *for* loop includes a *while* loop and two *for* loops. The computation complexity of the *while* loop (Lines 24–29) is $O(F_{\max}\theta_{\max}|I|)$. The computation complexity of the first *for* loop (Lines 4–12) and the second *for* loop (Lines 13–22) are $O(F_{\max}\theta_{\max})$ and $O(F_{\max}\theta_{\max}|I|)$, respectively. Thus, the running time of the *MD assignment* is $O(F_{\max}\theta_{\max}|I|)$. □

Lemma 4. *The winning BS selection is computationally efficient.*

Proof. In each iteration, the outer *while* loop includes two *for* loops and a minimization operation. The computation complexity of the first *for* loop (Lines 3–10) and the second *for* loop (Lines 15–17) are $O(F_{\max}\theta_{\max}|I||J|)$ and $O(|J|)$, respectively. The computation complexity of the minimization operation (Line 11) is $O(|J|)$. Since the outer loop is run no more than $|I|$ times, the running time of the *winning BS selection* is $O(F_{\max}\theta_{\max}|I|^2|J|)$. □

Lemma 5. *The critical-value-based charging is computationally efficient.*

Proof. The outer *for* loop is run no more than $|J|$ times. The middle *while* loop is run no more than $|I|$ times. The middle *while* loop includes two *for* loops and a sorting operation. The first *for* loop (Lines 5–12) needs the polynomial time $O(F_{\max}\theta_{\max}|I||J|)$. The sorting operation (Line 13) needs the polynomial time $O(|J|)$. The second *for* loop (Lines 15–17) needs the polynomial time $O(J)$. Consequently, the *critical-value-based charging* is computationally efficient, with the polynomial time $O(F_{\max}\theta_{\max}|I|^2|J|^2)$. □

Theorem 3. *TRAIM achieves computational efficiency.*

Proof. According to Lemmas 3, 4 and 5, each algorithm in TRAIM is computationally efficient. Consequently, TRAIM achieves computational efficiency. □

5. Performance evaluation

5.1. Simulation settings

We conduct the simulation studies according to the widely adopted EUA dataset [22], which includes the positions of each MD and each BS in Metropolitan Melbourne. In the simulations, we consider an MEC system that offers task offloading service in a region of $1000 \times 1000\text{m}^2$, in which 40 BSs and 180 MDs are deployed, and their positions are derived from the EUA dataset. For each MD, the required number of CPU cores is randomly generated in $[1, 5]$, and the required data transmission rate is randomly generated in $[1, 10]$ Mbps. For each BS, the available number of CPU cores is randomly generated within $[16, 24]$, the available number of orthogonal wireless subchannels is randomly picked from $[15, 30]$, each of which owns the bandwidth of 1 Mbps, and its maximum coverage radius is randomly chosen within $[100, 350]$ m. Moreover, the cost of per CPU core is randomly generated within $(0, 1]$, and the cost of per subchannel is also randomly generated within $(0, 1]$. Thus, the claimed cost of BS j for its resource combination, i.e., b_j, is uniformly generated within $(0, F_j + \theta_j]$. In addition, the background noise and the transmission power are randomly produced from $[-100, -50]$ dBm and $[200, 300]$mW [23],[24]. According to the physical interference model for wireless communications, we set the channel gain σ_{ij} as $D_{ij}^{-\psi}$ and set the path loss factor ψ as 4, where D_{ij} denotes the distance from MD i to BS j [23],[24]. Each simulation result is averaged by 50 times. The main simulation parameters are summarized in Table 2.

Table 2
Simulation parameters.

Parameter	Configuration
service region	$1000 \times 1000\text{m}^2$
number of BSs	40
number of MDs	180
required number of CPU cores	$[1, 5]$
required data transmission rate	$[1, 10]$Mbps
available number of CPU cores	$[16, 24]$
available number of orthogonal wireless subchannels	$[15, 30]$
bandwidth of each subchannel	1Mbps
maximum coverage radius	$[100, 350]$m
cost of per CPU core	$(0, 1])$
cost of per subchannel	$(0, 1])$
claimed cost of BS j	$(0, F_j + \theta_j]$
background noise	$[-100, -50]$dBm
transmission power	$[200, 300]$mW

For comparison purpose, we choose the following three auction-based mechanisms as baselines. 1) Random winner selection mechanism (RWSM): In each iteration, MDs are first randomly assigned to BSs, and then one BS is randomly selected as the winner. This process is iterated until all tasks have been offloaded. 2) Greedy winner selection mechanism (GWSM): In each iteration, it first adopts our proposed dynamic programming based algorithm to acquire the set of optimal BS-MD associations, and then it selects the BS with lowest cost as the winner. This process is iterated until all tasks have been offloaded. 3) RACORAM [33]: In this mechanism, a greedy randomized adaptive search procedure based winning bid scheduling method is proposed to decide the winning SBSs, and a payment rule based on the standard Vickrey-Clarke-Groves (VCG) scheme is designed to calculate the payments of SBSs. 4) Optimal mechanism (OPT): In this mechanism, the RAPTO is resolved in an optimal fashion, which is computationally infeasible when the size of MEC system grows. For charges, it uses a VCG-based charging strategy for BSs.

To conduct a comprehensive performance evaluation, we employ social cost (see Definition 2), overpayment ratio [25],[26] (see Definition 8), truthfulness (see Definition 3), individual rationality (see Definition 4) and running time as metrics. Here, the overpayment is the total payments for the winning BSs minus the social cost.

Definition 8 (Overpayment Ratio). The overpayment ratio is defined as the proportion of the total overpayment relative to the social cost, i.e.,

$$\lambda = \frac{P - H}{H} \tag{10}$$

where P and H denote the total payments and the social cost, respectively.

The overpayment ratio reflects the efficiency of the payment mechanism in a reverse auction. A lower overpayment ratio indicates that the platform's payments are closer to the BSs' actual costs, achieving higher payment efficiency and better resource allocation.

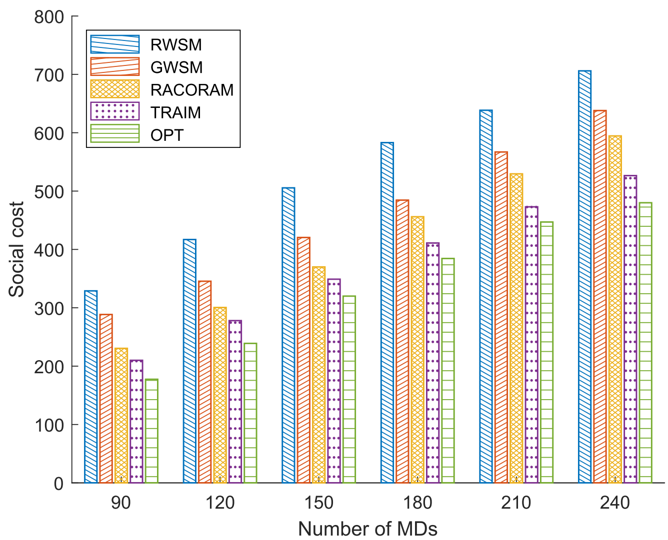

Fig. 2. Social cost with changed number of MDs.

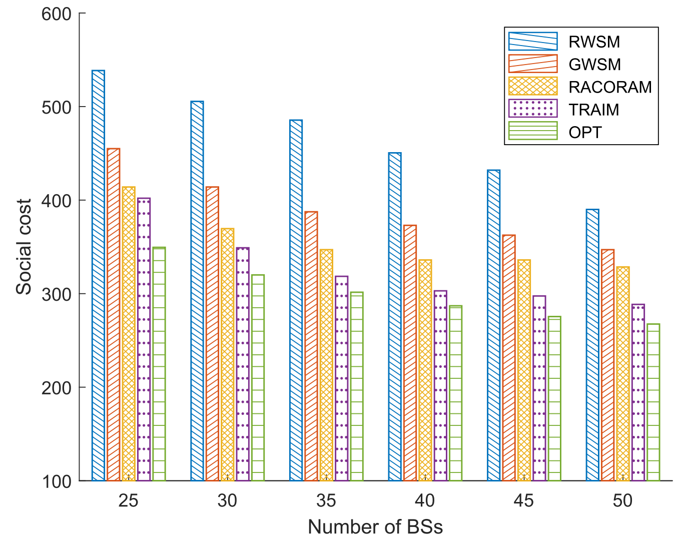

Fig. 3. Social cost with changed number of BSs.

5.2. Social cost

We evaluate our proposed TRAIM against RWSM, GWSM, RACORAM and OPT regarding social cost in various scenarios.

Fig. 2 illustrates the social cost of RWSM, GWSM, RACORAM, TRAIM and OPT when the number of MDs changes from 90 to 240. From Fig. 2, it can be observed that compared to RWSM and GWSM, TRAIM acquire lower social cost in the task offloading process. RACORAM also outperforms RWSM and GWSM, but still worse than TRAIM. That is because TRAIM employs the dynamic programming based approach to acquire the set of optimal BS-MD associations in each BS's coverage region, thus TRAIM has higher possibility to choose more suitable BSs to offload tasks in the process of the auction. Meanwhile, RACORAM may fall into a local optimum. As the number of MDs increases, more tasks need to be offloaded to BSs, causing the social cost of each involved mechanism will grow continuously.

Fig. 3 illustrates the social cost of RWSM, GWSM, RACORAM, TRAIM and OPT when the number of BSs changes from 25 to 50. From Fig. 3, we can see that TRAIM indicates an obvious superiority to RWSM, GWSM and RACORAM because the social cost obtained from TRAIM is always lower than that of RWSM, GWSM and RACORAM. This is because in TRAIM, dynamic MD assignment allows the platform to find the cheaper resources offered by BSs to execute MDs' tasks. In comparison with OPT that works in the optimal manner, the performance of TRAIM is only slightly worse, actually, it's very close.

Fig. 4 illustrates the social cost of RWSM, GWSM, RACORAM, TRAIM and OPT when the coverage radiuses changes from 100 to 350. From Fig. 4, it can be found that TRAIM still can acquire lower social cost than that of RWSM, GWSM and RACORAM. That is because with the increase of the coverage radius of each BS, more MDs are in its coverage region, which results in TRAIM can obtain better BS-MD association set in each BS's coverage region, and further have higher possibility to choose more cost-efficient BSs to offload tasks in the process of the auction. It is obvious that RWSM performs worst. That is because RWSM randomly chooses a set of BSs to take part in the task offloading.

Fig. 5 illustrates the social cost of RWSM, GWSM, RACORAM, TRAIM and OPT when the service regions change from 600×600 to 1400×1400. From Fig. 5, it can be found that TRAIM still can acquire lower social cost than that of RWSM, GWSM and RACORAM. In addition, as the service region increases, there are fewer MDs in the coverage of each BS. Thus, more BSs are needed to offload these MDs' task, causing the social cost of each involved mechanism will grow continuously.

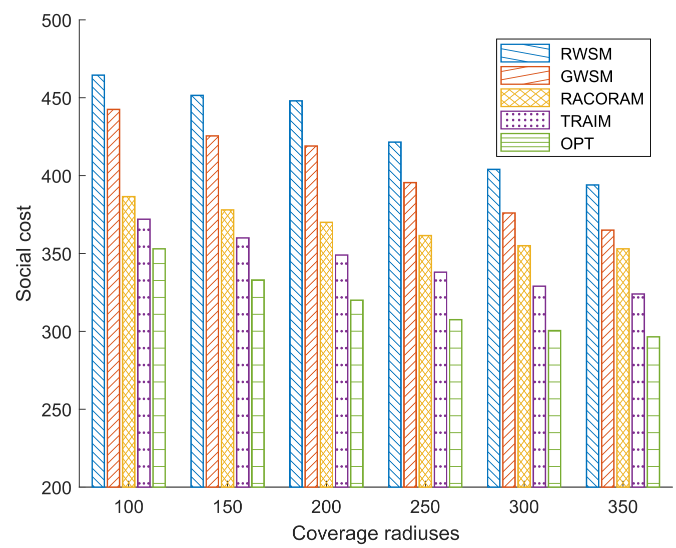

Fig. 4. Social cost with changed coverage radiuses.

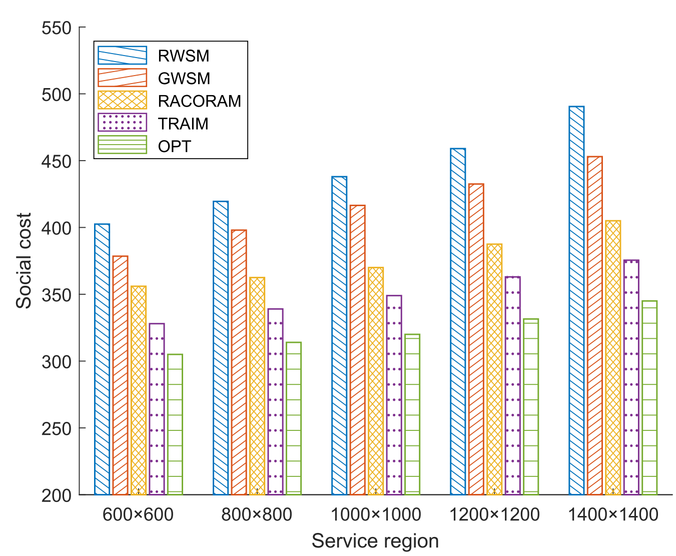

Fig. 5. Social cost with changed service regions.

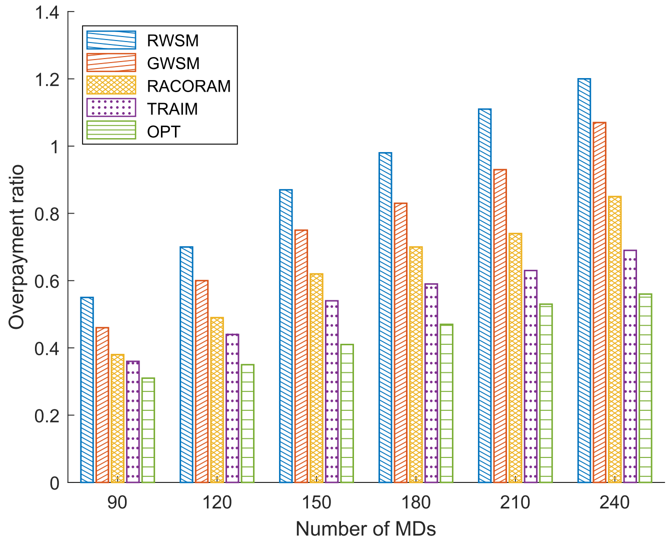

Fig. 6. Overpayment ratio with changed number of MDs.

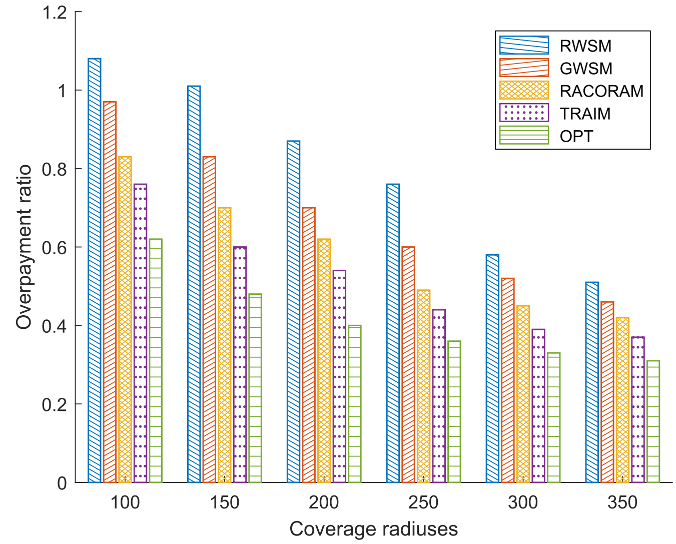

Fig. 8. Overpayment ratio with changed coverage radiuses.

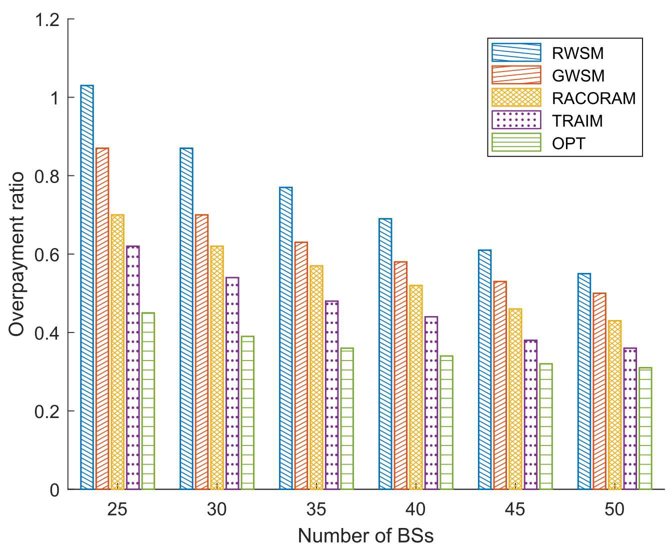

Fig. 7. Overpayment ratio with changed number of BSs.

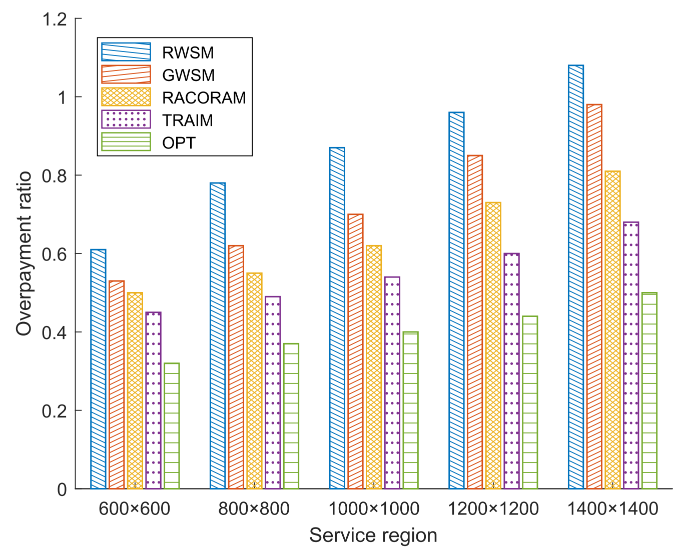

Fig. 9. Overpayment ratio with changed service regions.

5.3. Overpayment ratio

We compare our proposed TRAIM with RWSM, GWSM, RACORAM and OPT in terms of overpayment ratio under different scenarios.

We show the overpayment ratio of RWSM, GWSM, RACORAM, TRAIM and OPT in Fig. 6 as the number of MDs changes from 90 to 240. As shown in Fig. 6, TRAIM always performs much better than RWSM, GWSM and RACORAM as the number of MDs rises. That is because the real cost of the critical bid obtained by TRAIM is relatively smaller. Furthermore, it can also be observed that the overpayment ratio increases as the number of MDs grows, since there are more tasks to be offloaded for each BS, which leads to the real costs of the critical bids become higher.

We show the overpayment ratio of RWSM, GWSM, RACORAM, TRAIM and OPT in Fig. 7 as the number of BSs changes from 25 to 50. As shown in Fig. 7, TRAIM can obtain a smaller overpayment ratio than that of RWSM, GWSM and RACORAM. The reason is that TRAIM can get the optimal BS-MD association set for each BS through dynamic MD assignment and accept the cost-efficient bids as winning bids, which further results in the real costs of its critical bids being smaller. We can also observe that the overpayment ratio decreases when the number of BSs grows because there is a more fierce competition among BSs, further resulting in the real cost of the critical bid becoming lower.

We show the overpayment ratio of RWSM, GWSM, RACORAM, TRAIM and OPT in Fig. 8 as the coverage radiuses of BSs changes from 100 to 350. From Fig. 8, it can be found that TRAIM still can obtain smaller overpayment ratio than that of RWSM, GWSM and RACORAM. The main reason is that with the increasing of the coverage radiuses of BSs, more MDs are in their coverage regions, which results in TRAIM obtaining better BS-MD association set for each BS, and further increasing the possibility to choose the critical bids whose real costs is smaller. At the same time, it also can be found that the overpayment ratio of each involved mechanism diminishes as the coverage radiuses of BSs becomes larger. The main reason is that more and more tasks to be offloaded for each BS, which leads to the lower overpayment ratio.

Fig. 9 illustrates the overpayment ratio of RWSM, GWSM, RACORAM, TRAIM and OPT when the service regions change from 600×600 to 1400×1400. From Fig. 9, it can be found that TRAIM still can obtain smaller overpayment ratio than that of RWSM, GWSM and RACORAM. In addition, as the service region increases, there are fewer MDs in the coverage of each BS. Thus, more BSs are needed to offload these MDs'

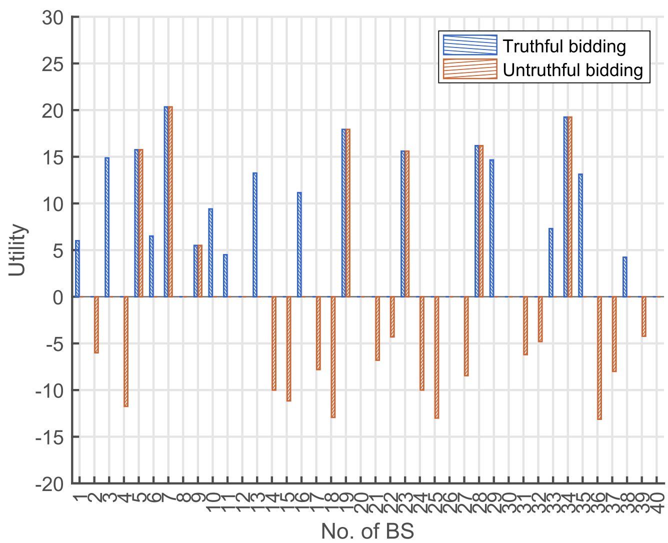

Fig. 10. Truthfulness.

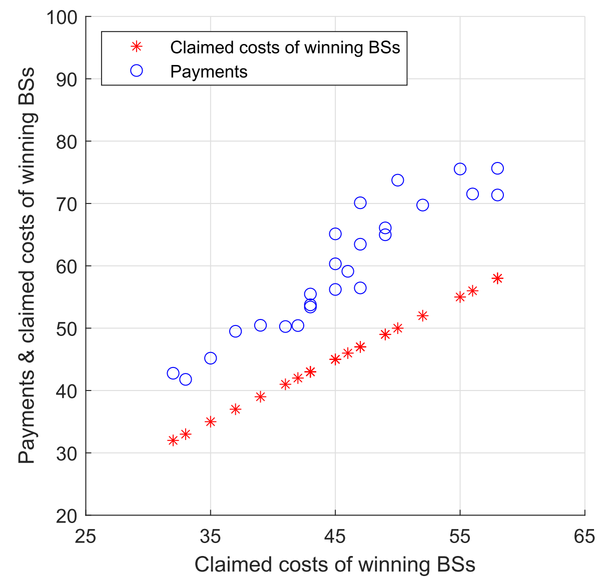

Fig. 11. Individual rationality.

task, causing the overpayment ratio of each involved mechanism will grow continuously.

5.4. Truthfulness

To verify the property of truthfulness in TRAIM, we select a set of BSs in one auction round, to examine how their utilities change when they respectively report truthful bids and untruthful bids. The corresponding results are shown in Fig. 10.

Fig. 10 shows for the winning BSs, if they report their true costs, they will acquire a non-negative utility and if they report their false costs, they will obtain the same utility or a zero utility. For the losing BSs, if they report their true costs, they will acquire a zero utility and if they report their false costs, they will obtain a zero utility or a negative utility. According to the above analyses, it can be found that TRAIM achieves truthfulness of BSs, because they cannot increase the utilities through untruthful reporting.

5.5. Individual rationality

To verify that TRAIM is individually rational, for each winning BS, we plot its claimed cost and the corresponding payment in Fig. 11. It can be found that the claimed cost of each winning BS is no more than its corresponding payment, i.e., the utilities of BSs are non-negative. Consequently, TRAIM is individually rational. This way, BSs are encouraged to supply their resources to execute MDs' task, in order to obtain a certain amount of compensation.

5.6. Computational efficiency

To verify TRAIM is computationally efficient, we measure its running time across various system scales, with the results illustrated in Table 3. For each system scale, the test runs 20 times, with average results evaluating computational efficiency.

From Table 3, it can be found that the running time of TRAIM gradually increases when the system scale grows. Here I and J denote the number of MDs and the number of BSs, respectively. What is more, the running time doesn't grow exponentially with the system scale. This conforms to Theorem 3, i.e., TRAIM achieves computational efficiency. In addition, it can be found that TRAIM needs more running time than RWSM and GWSM. Though TRAIM needs more running time, TRAIM enables to calculate the set of optimal MD-BS associations and selects the more cost-efficient bids, which further brings lower social cost and overpayment ratio.

Table 3
Running times (ms).

I/J	RWSM	GWSM	RACORAM	TRAIM	OPT
30/15	1.36×10^1	2.11×10^1	2.28×10^1	2.52×10^1	9.05×10^1
60/20	1.55×10^1	2.23×10^1	2.57×10^1	2.98×10^1	2.17×10^2
90/25	1.71×10^1	2.48×10^1	2.76×10^1	3.97×10^1	4.18×10^2
120/30	1.96×10^1	2.67×10^1	3.32×10^1	5.02×10^1	8.26×10^2
150/35	2.10×10^1	3.08×10^1	3.82×10^1	6.52×10^1	2.16×10^3
180/40	2.14×10^1	4.00×10^1	5.15×10^1	8.80×10^1	4.53×10^3
210/45	2.26×10^1	5.60×10^1	7.01×10^1	1.15×10^2	8.11×10^3
240/50	2.37×10^1	8.48×10^1	9.64×10^1	1.52×10^2	1.32×10^4

6. Related work

Emerging MEC enables MDs to offload the computation-intensive tasks to nearby edge servers. It is vitally important to efficiently utilize the limited resources of these edge servers. Recently, numerous auction-based mechanisms [27]–[39] have been deliberately developed for MEC.

Habiba et al. [27] design a task offloading service market using a repeated auction, in which multiple resource providers compete to offer the computation resources to offload tasks. Bahreini et al. [28] propose two novel resource allocation mechanisms, which consider characteristics of both combinatorial and position auctions and focus on users' heterogeneous requests. Huang et al. [29] present a reverse auction-based task assignment and urgency-value based transmission scheduling mechanism, whose objective is to maximize the platform's revenue while meeting the delay constraints of tasks. However, the mechanisms in [27]–[29] fail to guarantee truthfulness of MDs, which can not only harm other MDs, but also indirectly result in profit losses for the network operator.

Su et al. [30] propose a truthful combinatorial auction-based mechanism, which consists of resource allocation, buyer-seller matching, and payment calculation. He et al. [31] design an auction-based incentive mechanism, which can guarantee the desirable properties, such as truthfulness, individual rationality, and computational efficiency. Chen et al. [32] propose a truthful service-oriented double auction-based mechanism, which jointly considers the issues of service provisioning and incentive design in an MEC system. Zhou et al. [33] present a reverse auction-based mechanism to offload tasks and allocate resources, whose objective is to minimize the cloud service center's cost. Lin et al. [34] propose a truthful double auction-based mechanism with latency

Table 4
Comparison of TRAIM with existing auction-based mechanisms.

Ref.	RA	JCC	MMMB	T	LC	DRA	SCM
[27]	×	×	✓	×	×	✓	×
[28]	×	×	✓	×	×	✓	×
[29]	✓	✓	✓	×	×	×	✓
[30]	×	✓	✓	✓	×	✓	×
[31]	×	×	✓	✓	×	✓	×
[32]	×	✓	✓	✓	×	✓	×
[33]	✓	✓	✓	✓	×	✓	✓
[34]	×	✓	✓	✓	×	×	×
[35]	×	✓	✓	✓	×	✓	×
[36]	×	✓	✓	✓	×	✓	×
[37]	×	×	✓	✓	✓	✓	×
[38]	×	×	✓	✓	✓	✓	×
[39]	×	✓	✓	✓	✓	✓	×
TRAIM	✓	✓	✓	✓	✓	✓	✓

Reverse auction (RA), joint computation and communication resource allocation (JCC), multiple MDs and multiple BSs (MMMB), truthfulness (T), locality constraint (LC), dynamic resource allocation (DRA), social cost minimization (SCM).

guaranteed, in which resources in edge servers can be effectively allocated to offload MDs' tasks. Wang et al. [35] propose an online auction-based mechanism to facilitate the resource trading between edge clouds and MDs, aiming to achieve profit maximization. Zhang et al. [36] propose an online auction-based mechanism to maximize the long-term sum-of-rewards in the process of task offloading, while explicitly considering the highly dynamic nature of task arrivals. Nevertheless, the mechanisms in [31]–[36] overlook the locality constraints of an MEC system, where each MD can just offload its task to the nearby edge servers and each edge server can just serve MDs in its coverage region.

Sun et al. [37] present a double auction-based dynamic pricing mechanism to match MDs with BSs, aiming to pursue high system efficiency. Ma et al. [38] design a padding-based double auction mechanism to acquire the near-optimal solution, aiming to achieve social welfare maximization. Nevertheless, the mechanisms in [37],[38] just take into account the computation resources and neglect the communication resources, which is an essential resource for task offloading in an MEC system.

Wang et al. [39] propose a novel auction-based mechanism for resource allocation enabling flexible task offloading, aiming to achieve social welfare maximization while guaranteeing truthfulness under the locality constraints. However, the mechanism in [39] cannot be directly applied to our reverse auction scenario, where the bidders (as sellers) offer items to the user (as buyer).

To better understand of the differences between the proposed TRAIM and the existing auction-based mechanisms, a comparative study is presented in Table 4. Nevertheless, most of the existing auction-based mechanisms for MEC task offloading do not jointly consider the locality constraints in the reverse auction scenario as shown in Table 3. Furthermore, most of them do not consider the joint computation and communication resource allocation in the process of task offloading. Therefore, it is essential to develop effective mechanisms to fill the gap between the current approaches and the MEC task offloading.

7. Conclusion

In this paper, we devise a truthful reverse auction-based incentive mechanism for MEC task offloading called TRAIM. In TRAIM, we explicitly consider of the locality constraint, overlapped coverage and capacity constraint in process of MEC task offloading. To the specific, we first devise the *MD assignment* based on the dynamic programming approach to obtain the optimal BS-MD association set. Using this algorithm as a building block, we further design the *winning BS selection* based on the greedy method to decide the collection of winning BSs within polynomial time. Next, we compute the charge of each winning BS using the *critical-value-based charging*. Rigorous theoretical analysis and extensive simulations demonstrate the effectiveness of our proposed TRAIM. In the future, we plan to consider the protection of MDs' private information (e.g., the location and the bid), which can be used to infer MDs' hobbies or preferences. Further, we will adopt the cryptographic methods, including homomorphic encryption, order-preserving encryption, and garbled circuits, to protect the location and the bid information.

CRediT authorship contribution statement

Xueyi Wang: Writing – review & editing, Writing – original draft, Methodology; **Yi Ma:** Methodology, Formal analysis, Data curation; **Zhihao Qin:** Visualization, Software, Data curation, Validation; **Jian Xu:** Supervision, Methodology, Investigation; **Jianzhe Zhao:** Resources, Project administration, Data curation, Conceptualization; **Rongfei Zeng:** Writing – review & editing, Methodology; **Yang Song:** Validation, Investigation, Formal analysis; **Qiang He:** Validation, Project administration, Methodology.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] W. Xu, Y. Shen, N. Bergmann, W. Hu, Sensor-assisted multi-view face recognition system on smart glass, IEEE Trans. Mobile Comput. 17 (1) (2018) 197–210.
[2] X. Wen, S. Zhao, H. Wang, R. Wu, M. Qu, T. Hu, G. Chen, J. Tao, C. Fan, Multi-source multi-label learning for user profiling in online games, IEEE Trans. Multimedia 25 (2023) 4135–4147.
[3] A. Genay, A. Lécuyer, M. Hachet, Being an avatar "for real": a survey on virtual embodiment in augmented reality, IEEE Trans. Vis. Comput. Graph 28 (12) (2022) 5071–5090.
[4] M. Tang, V.W.S. Wong, Deep reinforcement learning for task offloading in mobile edge computing systems, IEEE. Trans. Mob. Comput. 21 (6) (2022) 1985–1997.
[5] P. Mach, Z. Becvar, Mobile edge computing: a survey on architecture and computation offloading, IEEE Commun. Surv. Tutor 19 (3) (2017) 1628–1656.
[6] M. Gao, R. Shen, L. Shi, W. Qi, J. Li, Y. Li, Task partitioning and offloading in DNN-task enabled mobile edge computing networks, IEEE. Trans. Mob. Comput. 22 (4) (2023) 2435–2445.
[7] Z. Cai, Z. Duan, W. Li, Exploiting multi-dimensional task diversity in distributed auctions for mobile crowdsensing, IEEE. Trans. Mob. Comput. 20 (8) (2021) 2576–2591.
[8] A.Y. Kiani, S.A. Hassan, B. Su, H. Pervaiz, Q. Ni, Minimizing the transaction time difference for NOMA-based mobile edge computing, IEEE Commun. Lett. 24 (4) (2020) 853–857.
[9] D. Li, Q. Yang, W. Yu, D. An, Y. Zhang, W. Zhao, Towards differential privacy-based online double auction for smart grid, IEEE Trans. Inf. Forensic Secur. 15 (2020) 971–986.
[10] X. Wang, L. Ma, X. Wang, Y. Shi, B. Yi, M. Huang, Truthful VNFI procurement mechanisms with flexible resource provisioning in NFV markets, IEEE Trans. Cloud Comput 11 (2) (2023) 1707–1718.
[11] H. Qiu, K. Zhu, N.C. Luong, C. Yi, D. Niyato, D.I. Kim, Applications of auction and mechanism design in edge computing: a survey, IEEE Trans. Cogn. Commun. Netw. 8 (2) (2022) 1034–1058.
[12] X. Wang, X. Wang, C. Wang, R. Zeng, L. Ma, Q. He, M. Huang, Truthful online combinatorial auction-based mechanisms for task offloading in mobile edge computing, IEEE. Trans. Mob. Comput. 24 (7) (2025) 6488–6502.
[13] S. Guo, Y. Dai, S. Guo, X. Qiu, F. Qi, Blockchain meets edge computing: stackelberg game and double auction based task offloading for mobile blockchain, IEEE Trans. Veh. Technol. 69 (5) (2020) 5549–5561.

[14] X. Wang, D. Wu, X. Wang, R. Zeng, L. Ma, R. Yu, Truthful auction-based resource allocation mechanisms with flexible task offloading in mobile edge computing, IEEE. Trans. Mob. Comput. 23 (5) (2024) 6377–6391.
[15] L. Ausubel, P. Milgrom, The lovely but lonely vickrey auction, in: Combinatorial Auctions, Cambridge, MA, USA, MIT Press, 2006.
[16] M.J. Osborne, A. Rubinstein, A Course in Game Theory, MIT Press, Cambridge, MA, USA, Cambridge, MA, USA, 1994.
[17] X. Wang, X. Wang, R. Zeng, L. Yan, D. Wu, Q. He, M. Huang, Truthful padding-based auction mechanisms for cross-cloud link bandwidth allocation and pricing, https://doi.org/10.1109/TON.2025.3582304
[18] D. Zhang, L. Tan, J. Ren, M. Awad, S. Zhang, Y. Zhang, P. Wan, Near-optimal and truthful online auction for computation offloading in green edge-computing systems, IEEE. Trans. Mob. Comput. 19 (4) (2020) 880–893.
[19] W. Sun, J. Liu, Y. Yue, P. Wang, Joint resource allocation and incentive design for blockchain-based mobile edge computing, IEEE Trans. Wirel. Commun. 19 (9) (2020) 6050–6064.
[20] N. Nisan, Algorithmic Game Theory, Cambridge University Press, 2007.
[21] A. Mu, N. Nisan, Truthful approximation mechanisms for restricted combinatorial auctions, Games Econ. Behav. 64 (2) (2008) 612–631.
[22] Dataset, 2024. https://github.com/swinedge/eua-dataset.
[23] X. Chen, W. Li, S. Lu, Z. Zhou, X. Fu, Efficient resource allocation for on-demand mobile-edge cloud computing, IEEE Trans. Veh. Technol. 67 (9) (2018) 8769–8780.
[24] X. Chen, L. Jiao, W. Li, X. Fu, Efficient multi-user computation offloading for mobile-edge cloud computing, IEEE/ACM Trans. Netw. 24 (5) (2016) 2795–2808.
[25] Z. Feng, Y. Zhu, Q. Zhang, L.M. Ni, A.V. Vasilakos, TRAC: truthful auction for location-aware collaborative sensing in mobile crowdsourcing, in: Proc. of IEEE INFOCOM, of IEEE INFOCOM, 2014.
[26] H. Cai, Y. Zhu, Z. Feng, A truthful incentive mechanism for mobile crowd sensing with location-Sensitive weighted tasks, Comput. Netw. 132 (2018) 1–14.
[27] U. Habiba, S. Maghsudi, E. Hossain, A repeated auction model for load-aware dynamic resource allocation in multi-access edge computing, IEEE Trans. Mobile Comput. 23 (7) (2024) 7801–7817.
[28] T. Bahreini, H. Badri, D. Grosu, Mechanisms for resource allocation and pricing in mobile edge computing systems, IEEE Trans. Parallel Distrib. Syst. 33 (3) (2022) 667–682.
[29] X. Huang, G. Ji, B. Zhang, C. Li, Platform profit maximization in D2D collaboration based multi-access edge computing, IEEE Trans. Wirel. Commun. 22 (7) (2023) 4282–4295.
[30] Y. Su, W. Fan, Y. Liu, F. Wu, A truthful combinatorial auction mechanism towards mobile edge computing in industrial internet of things, IEEE Trans. Cloud Comput. 11 (2) (2023) 1678–1691.
[31] J. He, D. Zhang, Y. Zhou, Y. Zhang, A truthful online mechanism for collaborative computation offloading in mobile edge computing, IEEE Trans. Ind. Inform. 16 (7) (2020) 4832–4841.
[32] X. Chen, G. Zhu, H. Ding, L. Zhang, H. Zhang, Y. Fang, End-to-end service auction: a general double auction mechanism for edge computing services, IEEE/ACM Trans. Netw. 30 (6) (2022) 2616–2629.
[33] H. Zhou, T. Wu, X. Chen, S. He, D. Guo, J. Wu, Reverse auction-based computation offloading and resource allocation in mobile cloud-edge computing, IEEE Trans. Mob. Comput. 22 (10) (2023) 6144–6159.
[34] J. Lin, L. Huang, H. Zhang, X. Yang, P. Zhao, A novel latency guaranteed based resource double auction for market-oriented edge computing, Comput. Netw. 189 (2021).
[35] Q. Wang, S. Guo, J. Liu, C. Pan, L. Yang, Profit maximization incentive mechanism for resource providers in mobile edge computing, IEEE Trans. Serv. Comput. 15 (1) (2022) 138–149.
[36] D. Zhang, L. Tan, J. Ren, M. Awad, S. Zhang, Y. Zhang, P. Wan, Near-optimal and truthful online auction for computation offloading in green edge-computing systems, IEEE. Trans. Mobile Comput. 19 (4) (2020) 880–893.
[37] W. Sun, J. Liu, Y. Yue, H. Zhang, Double auction-based resource allocation for mobile edge computing in industrial internet of things, IEEE Trans. Ind. Inform 14 (10) (2018) 4692–4701.
[38] L. Ma, X. Wang, X. Wang, L. Wang, Y. Shi, M. Huang, TCDA: truthful combinatorial double auctions for mobile edge computing in industrial internet of things, IEEE Trans. Mobile Comput. 21 (11) (2022) 4125–4138.
[39] X. Wang, D. Wu, X. Wang, R. Zeng, L. Ma, R. Yu, Truthful auction-based resource allocation mechanisms with flexible task offloading in mobile edge computing, IEEE Trans. Mobile Comput. 23 (5) (2024) 6377–6391.